Shift-Uniform Combes–Thomas Resolvents on Physical Gauge Cochains: A Lean-Verified Route to Exponentially Localized Covariance Roots

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Abstract

Let K be a coercive finite-range operator on the finite-dimensional real Hilbert space of physical lattice gauge one-cochains. A standard Combes–Thomas argument controls the inverse kernel by conjugating K with an exponential weight. Applying that argument naively to the shifted family $K + s\mathrm{Id}$ appears to enlarge the entrywise bound by s, destroying the uniformity needed to integrate resolvents. We isolate and machine-check the missing algebraic fact: scalar identity shifts cancel exactly from the tilt defect,

$$(K + s\mathrm{Id})_\theta - (K + s\mathrm{Id}) = K_\theta - K.$$

Consequently, one fixed Combes–Thomas budget for K controls every $K + s\mathrm{Id}$, $s \geq 0$. If K is coercive with constant $c > 0$, has range R, single-bond kernel bound M, and ball-cardinality bound N_R, then under $M(e^{\theta R} - 1)N_R \leq c/2$, every right inverse C_s of $K + s\mathrm{Id}$ satisfies

$$\|(C_s)_{q,p}\|_{\mathrm{op}} \leq \frac{2}{c+s} e^{-\theta \operatorname{dist}(q,p)}.$$

The cancellation identity, shifted coercivity, uniform defect estimate, and resolvent kernel theorem are formalized in Lean 4 on the repository's actual physical gauge-cochain types. At paper level, self-adjoint positivity and the Stieltjes formula then give

$$\|(K^{-1/2})_{q,p}\|_{\mathrm{op}} \leq \frac{2}{\sqrt{c}} e^{-\theta \operatorname{dist}(q,p)}.$$

The fractional-power decay itself is classical in matrix analysis; the contribution claimed here is the shift-uniform, source-facing, machine-checked route with explicit constants. In version 0.4 the positive real continuous- linear-map root is also constructed and verified spectrally in Lean, and a new adapter derives every algebraic root-certificate field from covariance positivity. The remaining gap is stated without compression: the Stieltjes identity and integrated single-bond root-kernel bound are not yet formalized, so the localized physical root certificate remains conditional on that kernel estimate; fixed-volume coercivity is still not volume-uniform.

1 Claims and status discipline

This paper separates three levels that are easy to conflate.

*This manuscript and part of the formal development were produced with AI assistance under the repository's audit protocol. Claims labelled FORMALIZED were checked by Lean 4 against a pinned Mathlib and a targeted axiom oracle. Paper-level functional-calculus arguments are labelled PROVED and are not represented as Lean theorems.

Status	Meaning in this manuscript
FORMALIZED	A named Lean theorem compiles on the pinned tree and the targeted axiom oracle reports only `propext`, `Classical.choice`, and `Quot.sound`.
PROVED	A complete finite-dimensional mathematical proof is given here, but the proof is not yet a Lean declaration.
OPEN	The statement is neither assumed silently nor claimed. It is a named frontier for subsequent formalization or analysis.

The central resolvent result, Theorem 5.2, is FORMALIZED. The positive continuous-linear-map square root and its exact square, positivity, and symmetry properties are now also FORMALIZED. The Stieltjes passage to $K^{-1/2}$, Theorem 6.1, is PROVED at paper level. Its identification with the formal spectral root, the integrated single-bond kernel estimate, and unconditional discharge of the existing `PhysicalLocalizedCovarianceRootCertificate` remain OPEN. No continuum limit, mass gap, or volume-uniform positive coercivity constant is claimed.

The mathematical phenomenon is not presented as new. Exponential decay of inverses of banded matrices goes back at least to Demko–Moss–Smith [2]; decay bounds for completely monotone functions, including negative fractional powers, were developed directly by Benzi–Simoncini [3]. Our novelty claim is deliberately narrower: a machine-checked shift-cancellation mechanism and a reusable constant-explicit resolvent theorem on the physical gauge-cochain interface used by the surrounding formal development.

2 Physical cochains and localization predicates

Fix $d, N, N_c \in \mathbb{N}$ with $N > 0$. Let $\Lambda_N = (\mathbb{Z}/N\mathbb{Z})^d$ and let $\mathcal{B}_{d,N}$ be the finite set of positively oriented lattice bonds. The internal coordinate space is the real coordinate model of $\mathfrak{su}(N_c)$,

$$V_{N_c} \cong \mathbb{R}^{N_c^2-1}.$$

The finite-dimensional real Hilbert space of physical one-cochains is

$$\mathcal{H}_{d,N,N_c} := \ell^2(\mathcal{B}_{d,N}; V_{N_c}).$$

In Lean these are, respectively, `PhysicalBond d N`, `SUNLieCoord Nc`, and `PhysicalGaugeOneCochain d N Nc`.

For $p \in \mathcal{B}_{d,N}$ and $v \in V_{N_c}$, write $\delta_p v$ for the cochain supported at p with value v. This is `singlePhysicalBondCochain`. Let

$$\mathrm{dist} : \mathcal{B}_{d,N} \times \mathcal{B}_{d,N} \longrightarrow \mathbb{N}$$

be symmetric, satisfy the triangle inequality, and obey $\mathrm{dist}(p,p) = 0$.

Let $K : \mathcal{H} \to \mathcal{H}$ be a continuous linear map. The formal finite-range and kernel hypotheses have the following mathematical content:

$$\mathrm{dist}(q,p) > R \Longrightarrow (K\delta_p v)(q) = 0, \tag{1}$$
$$\|(K\delta_p v)(q)\| \le M\|v\|. \tag{2}$$

The cardinality parameter N_R controls every metric ball,

$$\#\{y : \operatorname{dist}(x, y) \le R\} \le N_R. \tag{3}$$

Finally, K is coercive with constant c if

$$c\|f\|^2 \le \langle f, Kf\rangle \qquad (f \in \mathcal{H}). \tag{4}$$

The corresponding Lean predicate is `IsCoerciveCLM K c`.

3 Exponential conjugation

Fix a root bond r and $\theta \in \mathbb{R}$. Define the diagonal tilt

$$(T_{\theta,r} f)(q) = e^{\theta \operatorname{dist}(r,q)} f(q) \tag{5}$$

and the conjugated operator

$$K_{\theta,r} = T_{\theta,r} K T_{-\theta,r}. \tag{6}$$

Because the bond set is finite, these are bounded maps without a domain issue. The formal development proves $T_{\theta,r}T_{-\theta,r} = \mathrm{Id}$ and, on a single-bond probe,

$$(K_{\theta,r}\delta_p v)(q) = e^{\theta(\operatorname{dist}(r,q)-\operatorname{dist}(r,p))}(K\delta_p v)(q). \tag{7}$$

On the range of K, the triangle inequality gives $|\operatorname{dist}(r, q) - \operatorname{dist}(r, p)| \le R$. The block Schur estimate in the formal library then yields

$$\|K_{\theta,r} - K\| \le \Delta_\theta, \qquad \Delta_\theta := M(e^{\theta R} - 1)N_R, \quad \theta \ge 0. \tag{8}$$

This is the existing Combes–Thomas engine. The issue for a Stieltjes construction is whether the same Δ_θ controls all positive scalar shifts.

4 The scalar-shift cancellation

Lemma 4.1 (identity is fixed; FORMALIZED)**.** *For every r, θ, and physical gauge-cochain space,*

$$T_{\theta,r}\mathrm{Id}T_{-\theta,r} = \mathrm{Id}.$$

In Lean this is `physicalTiltConjCLM_id`*.*

Proof. It is the inverse relation $T_{\theta,r}T_{-\theta,r} = \mathrm{Id}$. The Lean proof unfolds the conjugation and rewrites with the already formalized composition identity. □

Lemma 4.2 (shift cancellation; FORMALIZED)**.** *For $s \in \mathbb{R}$,*

$$(K + s\mathrm{Id})_{\theta,r} - (K + s\mathrm{Id}) = K_{\theta,r} - K. \tag{9}$$

In Lean this is `physicalTiltConjCLM_add_smul_id_sub`*.*

Proof. Conjugation is linear in the operator. Lemma 4.1 gives $(K+s\mathrm{Id})_{\theta,r} = K_{\theta,r}+s\mathrm{Id}$; subtracting $K + s\mathrm{Id}$ proves (9). The Lean proof separately establishes additivity and scalar compatibility of conjugation and closes the operator identity by abelian-group normalization. □

The importance of (9) is not its algebraic difficulty but its location in the estimate. A naive application of the unshifted theorem to $K + s\mathrm{Id}$ asks for a kernel bound on the whole shifted operator and therefore introduces a spurious term proportional to s. Equation (9) shows that the tilt only sees the off-diagonal commutator; the scalar diagonal shift is invisible.

Corollary 4.3 (uniform defect; FORMALIZED). *Under (1)–(3), for every $s \in \mathbb{R}$,*

$$\|(K + s\mathrm{Id})_{\theta,r} - (K + s\mathrm{Id})\| \leq \Delta_\theta.$$

This is `norm_physicalTiltConj_add_smul_id_sub_le`.

5 Shift-uniform coercivity and resolvents

The scalar shift also improves coercivity exactly.

Lemma 5.1 (shifted coercivity; FORMALIZED). *If (4) holds, then*

$$(c + s)\|f\|^2 \leq \langle f, (K + s\mathrm{Id})f\rangle$$

for every real s. In Lean this is `isCoerciveCLM_add_smul_id`.

Combining Lemma 5.1 with Corollary 4.3 gives the shift-uniform CT2 endpoint

$$\langle f, (K + s\mathrm{Id})_{\theta,r} f\rangle \geq (c + s - \Delta_\theta)\|f\|^2. \tag{10}$$

This is `isCoerciveCLM_physicalTiltConj_add_smul_id`. If $s \geq 0$ and $\Delta_\theta \leq c/2$, then

$$c + s - \Delta_\theta \geq \frac{c + s}{2}. \tag{11}$$

The corresponding formal theorem is `isCoerciveCLM_physicalTiltConj_add_smul_id_half`.

Theorem 5.2 (shift-uniform Combes–Thomas resolvent; FORMALIZED). *Assume $c > 0$, $M \geq 0$, $\theta > 0$, equations (1)–(4), and*

$$M(e^{\theta R} - 1)N_R \leq \frac{c}{2}. \tag{12}$$

For $s \geq 0$, let C_s be any right inverse of $K + s\mathrm{Id}$:

$$(K + s\mathrm{Id})C_s = \mathrm{Id}.$$

Then, for all bonds p, q and $v \in V_{N_c}$,

$$\|(C_s \delta_p v)(q)\| \leq \frac{2}{c + s} e^{-\theta\,\mathrm{dist}(q,p)}\|v\|. \tag{13}$$

This is `physicalCovariance_exponentialKernelBound_of_coercive_add_smul_id`.

Proof architecture. Root the tilt at the source bond p and set $y = T_{\theta,p} C_s \delta_p v$. Since $\mathrm{dist}(p, p) = 0$, the source probe is fixed by its own tilt. The right-inverse relation and exact untilting give

$$(K + s\mathrm{Id})_{\theta,p} y = \delta_p v.$$

Coercivity at $(c+s)/2$ and Cauchy–Schwarz imply

$$\|y\| \le \frac{2}{c+s}\|v\|.$$

At the target bond,

$$(C_s\delta_p v)(q) = e^{-\theta\,\mathrm{dist}(p,q)}y(q).$$

The coordinate projection has norm at most one, and symmetry of the distance gives (13). Every equality and inequality in this proof architecture is represented in the cited Lean theorem. □

Remark 5.3 (why the denominator is integrable)**.** *The key improvement over a shift-blind estimate is not merely that the decay rate remains positive. The amplitude retains the spectral shift as* $(c+s)^{-1}$. *Substituting* $s = t^2$ *produces an integrable majorant on* $[0,\infty)$, *which is exactly what the inverse-square-root Stieltjes formula requires.*

6 The covariance-root corollary

The next theorem is ordinary finite-dimensional spectral calculus. It is included with a complete proof to identify the exact remaining Lean task.

Theorem 6.1 (localized inverse square root; Proved)**.** *In addition to the hypotheses of Theorem 5.2, assume that* K *is self-adjoint and positive with* $\sigma(K) \subset [c,\infty)$, $c > 0$. *Then the unique positive inverse square root* $K^{-1/2}$ *satisfies*

$$\|(K^{-1/2}\delta_p v)(q)\| \le \frac{2}{\sqrt{c}}e^{-\theta\,\mathrm{dist}(q,p)}\|v\|. \tag{14}$$

Moreover,

$$\|K^{-1/2}\| \le c^{-1/2}, \qquad K^{-1/2}K^{-1/2} = K^{-1}.$$

Proof. The spectral theorem gives the norm-convergent Stieltjes identity

$$K^{-1/2} = \frac{2}{\pi}\int_0^\infty (K + t^2\mathrm{Id})^{-1}\,dt. \tag{15}$$

Indeed, after diagonalizing K, the operator identity reduces to the scalar identity, valid for $\lambda > 0$,

$$\frac{2}{\pi}\int_0^\infty \frac{dt}{\lambda + t^2} = \frac{2}{\pi}\frac{\pi}{2\sqrt{\lambda}} = \lambda^{-1/2}. \tag{16}$$

There is no interchange difficulty in finite dimension. Apply Theorem 5.2 with $s = t^2$ and integrate the pointwise norm majorant:

$$\begin{aligned}\|(K^{-1/2}\delta_p v)(q)\| &\le \frac{2}{\pi}\int_0^\infty \frac{2}{c+t^2}e^{-\theta\,\mathrm{dist}(q,p)}\|v\|\,dt \\ &= \frac{2}{\sqrt{c}}e^{-\theta\,\mathrm{dist}(q,p)}\|v\|.\end{aligned}$$

The spectral bounds and the square identity follow by applying $\lambda \mapsto \lambda^{-1/2}$ on $\sigma(K)$. □

Remark 6.2 (constant accounting)**.** *The spectral norm of* $K^{-1/2}$ *is at most* $c^{-1/2}$, *whereas the kernel amplitude obtained from the half-budget CT argument is* $2c^{-1/2}$. *The factor two is the price of reserving half the coercivity uniformly for every tilt. We make no optimality claim for this constant or for the admissible rate* θ.

7 Connection to the formal covariance-root frontier

The repository already contains the proposition-valued structure `PhysicalLocalizedCovarianceRootCertificate`. Given a precision K, a covariance C, and a proposed root B, it requires:

1. a localized covariance certificate for (K, C);
2. $B \circ B = C$;
3. a norm bound on B;
4. self-adjointness of B in inner-product form;
5. positivity of B;
6. a single-bond kernel bound on B.

Theorem 6.1 supplies the mathematical content of items 2–6 when $C = K^{-1}$ and $B = K^{-1/2}$, while Theorem 5.2 supplies the uniform resolvent estimate needed for item 6. The new module `FiniteDimensionalRealPositiveSqrt.lean` now supplies a Lean term B for every positive finite-dimensional real continuous endomorphism and proves $B \circ B = C$, positivity, and symmetry. The physical adapter derives items 2–5 (using $\|B\|$ itself for the non-quantitative norm bound); item 6 is still an explicit analytic input.

The local-SPD route in `PhysicalGaugeLocalSPDPrecisionRoot.lean` is complementary. It develops Neumann and inverse-square-root coefficient majorants under a normalized small-perturbation hypothesis $K = a(\mathrm{Id} - L)$ with $\|L\| < 1$, but deliberately leaves the continuous-linear root and its kernel bound as source fields. The present route replaces that smallness hypothesis by coercivity plus finite range. Version 0.4 closes the positive-root object for every symmetric coercive precision; neither route yet proves the physical single-bond kernel bound for that canonical root.

The exact next formal chain is therefore

$$\boxed{\begin{array}{c}\text{self-adjoint coercive } K \longrightarrow \text{formal positive spectral root } B, \quad B^2 = K^{-1} \\ \longrightarrow \text{formal identification with the Stieltjes integral} \longrightarrow \text{integrated single-bond kernel bound} \\ \longrightarrow \texttt{PhysicalLocalizedCovarianceRootCertificate}.\end{array}} \tag{17}$$

The first arrow is now machine-checked by transporting a real positive- semidefinite matrix root through an orthonormal basis. The remaining engineering question is to identify this spectral root with the norm-convergent Stieltjes integral while preserving the single-bond kernel API.

Post-review implementation audit. The available library route was not a one-line application of an existing operator square-root theorem. The implementation chooses an orthonormal coordinate model, applies the real matrix spectral theorem, takes the nonnegative square root of every eigenvalue, and transports the result back to a continuous linear map. Lean verifies positive semidefiniteness and the exact square before transport, then verifies the continuous-linear-map properties after transport. This closes the first two obligations identified in version 0.3.1. The two remaining obligations are the norm-convergent Stieltjes identification and the integrated single-bond kernel estimate. Supplying the last predicate as an assumption still does not prove decay; the new certificate adapter uses it only to isolate the exact remaining frontier.

8 Prior art and exact delta

Combes and Thomas introduced exponential conjugation in spectral analysis [1]. Demko, Moss, and Smith proved exponential decay of inverse band matrices with spectral-rate information [2]. Benzi and Simoncini treated completely monotone functions of Hermitian banded and sparse matrices, explicitly including negative fractional powers [3]. Thus neither Theorem 6.1 nor the use of a resolvent integral is claimed as a new analytic phenomenon.

Antecedent	Overlap	Exact difference claimed here
Combes–Thomas [1]	Exponential weighting and resolvent localization	Finite physical gauge-cochain API, explicit Schur budget, exact scalar-shift cancellation, and machine-checked source-rooted extraction.
Demko–Moss–Smith [2]	Off-diagonal decay of inverses	Metric finite-range block operators rather than scalar band matrices; the main delta is formal verification, not a sharper rate.
Benzi–Simoncini [3]	Decay of negative fractional powers via matrix-function representations	No claim of analytic priority. We provide a Lean-verified resolvent family with the precise $(c+s)^{-1}$ amplitude consumed by the physical covariance-root interface.
Repository fixed-volume CT chain [8]	Formal inverse decay with amplitude $2/c$	The old theorem did not preserve a scalar shift; the new theorem is uniform for all $s \geq 0$ and enables the Stieltjes integral.
Repository local-SPD root frontier	Inverse-square-root coefficient and kernel majorants	Present hypotheses are coercivity plus finite range, not $\|L\| < 1$; the canonical real root is now formal, while its localized kernel bound remains open.

The literature search for this manuscript was performed on August 1, 2026, using combinations of “Combes–Thomas”, “inverse square root”, “fractional powers”, “sparse matrix functions”, “Stieltjes”, and “Lean”. We found no machine-checked Combes–Thomas inverse-square-root theorem, but this negative search result is not an exhaustiveness claim. The paper’s defensible dominance axis is reproducibility and integration into a concrete formal physics interface, not classical analytic novelty.

9 Adversarial audit and limitations

Attack 1: the theorem is already known under matrix-function language. Correct at the analytic level. This is why the title says “Lean-verified route” and not “new localization theorem”, and why Section 8 treats negative fractional powers as direct prior art.

Attack 2: the result is only finite volume. Correct. The formal physical cochain space is finite-dimensional. A separate machine-checked Poincaré-wall result shows that the current unscaled block-Poincaré route does not provide a positive coercivity constant uniform in volume for the relevant dimensions [8]. The present theorem preserves whatever c is supplied; it does not manufacture a uniform c.

Attack 3: coercivity does not imply a positive square root for a non-self-adjoint operator. Correct. The formal resolvent theorem does not need self-adjointness, but Theorem 6.1

adds self-adjoint positivity explicitly. Omitting it would be a false bridge.

Attack 4: a right inverse family is not a measurable or continuous resolvent map. In finite dimension the canonical inverse $(K + t^2\mathrm{Id})^{-1}$ is continuous in t for a positive self-adjoint K. The Lean theorem is intentionally pointwise in an arbitrary right inverse; the continuity and its integral identification with the now-formal spectral root belong to the open Stieltjes layer.

Attack 5: the diagonal is not genuinely off-diagonal data. The formal global kernel bound on the base operator still includes diagonal entries. The improvement is that the *shift* is eliminated exactly from the tilt defect. We do not claim a new off-diagonal norm predicate in this version.

Attack 6: the constants may be poor. Correct. The factor two and the Schur cardinality N_R are explicit but not optimized. Polynomial or best-approximation methods may give sharper rates for banded Hermitian matrices. The purpose here is a robust formal interface.

Kill criteria. The programme should be reformulated if any of the following occurs: (i) the exact shift identity fails under the actual root representation; (ii) the functional-calculus bridge cannot preserve the single-bond kernel API without assuming the desired kernel bound; (iii) the root certificate requires a volume-uniform c from the already refuted block normalization; or (iv) a prior formalization is found with the same physical types, hypotheses, constants, and endpoint. None is currently observed, but (ii) is the real technical bottleneck.

10 Lean theorem map and reproducibility

10.1 Theorem-to-artifact map

The resolvent source modules are `YangMills/RG/PhysicalCoerciveCombesThomas.lean` (CT1/CT2) and `YangMills/RG/PhysicalCoerciveCombesThomasInverse.lean` (CT3/CT4). The root layer is implemented in `FiniteDimensionalRealPositiveSqrt.lean`, `CoerciveCovariancePositiveSqrt.lean`, and `PhysicalGaugeCovariancePositiveRoot.lean`.

Lean declaration	File	Status
`physicalTiltConjCLM_id`	CT1/CT2 module	EXACT
`physicalTiltConjCLM_add_smul_id_sub`	CT1/CT2 module	EXACT
`isCoerciveCLM_add_smul_id`	CT1/CT2 module	EXACT
`norm_physicalTiltConj_add_smul_id_sub_le`	CT1/CT2 module	EXACT
`isCoerciveCLM_physicalTiltConj_add_smul_id`	CT1/CT2 module	EXACT
`isCoerciveCLM_physicalTiltConj_add_smul_id_half`	CT3/CT4 module	EXACT
`physicalCovariance_exponentialKernelBound_of_coercive_add_smul_id`	same	EXACT
`finiteDimensionalRealPositiveSqrt_comp_self`	`FiniteDimensionalRealPositiveSqrt.lean`	EXACT
`finiteDimensionalRealPositiveSqrt_isPositive`	same	EXACT

Lean declaration	File	Status
`covarianceSqrtOfIsCoerciveCLM_comp_self`	`CoerciveCovariancePositive Sqrt.lean`	EXACT
`physicalLocalizedCovariance RootCertificate_of_positive_covariance`	`PhysicalGaugeCovariance PositiveRoot.lean`	CONDITIONAL
Stieltjes identity and Theorem 6.1	this manuscript	paper
Integrated root-kernel bound	not yet implemented	open

10.2 Snapshot and commands

Repository: https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME. The shift-uniform resolvent artifact is the dedicated Git commit on branch `codex/stieltjes-root`

`b542b3f2f066b60a914977f2a50160e403a53367`.

Its parent is `81e9cf064d6771f986978ccffb96018c78b5e459`. The commit contains exactly the two theorem modules, the oracle registrations, and the verification-ledger entry. Thus the formal source cited here is an immutable Git object rather than an uncommitted working-tree patch. Remote publication and a hosted CI run of that commit remain release actions and are not claimed by this manuscript. The positive-root extension is the later immutable commit

`bd11e55f9cfe572d5ec4200f49c45339df4e3c03`.

It adds the three root modules, their aggregate imports, oracle registrations, and verification-ledger Addendum 571. The manuscript edits follow that commit, so the root source hash is not circular. The version-0.2 manuscript artifact is commit `25f532eaeecbfb66ee146e264953cde0e2e0ea29`. Version 0.3 is distributed under the annotated local release tag `shift-uniform-ct-v0.3` together with a complete Git source archive, rather than as a PDF alone. The archive contains this manuscript, the pinned `lake-manifest.json`, both Lean modules, `oracle_check.lean`, the focused `ShiftUniformOracle.lean`, and a verification transcript. A reader can therefore materialize and check the cited source without relying on an uncommitted worktree.

For cryptographic continuity, the materialized version-0.3 source package is the tree at commit `cba2b5d705bfadc377a9c04293a57a6bfcb82c19`. Its exact file name is `shift-uniform-combes-thomas-v0.3-source.zip`; it contains 7,350 archive entries and has SHA-256

`1B5EAB37038E071F1E3A1A1BA263D93048828EA079950273BF915B4605897823`.

Version 0.3.1 was the documentation-only revision that printed those package identifiers in the PDF. Version 0.4 is a scientific revision: it cites the separate positive-root commit above and updates the formal/open boundary. It does not alter the immutable version-0.3 source archive or its hash.

Toolchain: Lean `leanprover/lean4:v4.29.0-rc6`; Mathlib revision `0764272048...` (the full revision is recorded in `lake-manifest.json`). The focused checks are:

```
lake env lean YangMills/RG/PhysicalCoerciveCombesThomas.lean
lake env lean YangMills/RG/PhysicalCoerciveCombesThomasInverse.lean
lake build +YangMills.RG.PhysicalCoerciveCombesThomas:olean
lake build +YangMills.RG.PhysicalCoerciveCombesThomasInverse:olean
```

```
lake build YangMills.RG.FiniteDimensionalRealPositiveSqrt
lake build YangMills.RG.CoerciveCovariancePositiveSqrt
lake build YangMills.RG.PhysicalGaugeCovariancePositiveRoot
```

The four resolvent checks and all three root-module builds completed successfully on August 1, 2026. Two additional attempts to rebuild the full `YangMillsCore` aggregator were terminated by the command wrapper after 10 and 20 minutes, respectively, without a Lean error; no successful fresh full-core rebuild is claimed. A targeted oracle checked the seven resolvent declarations plus five root declarations; each printed exactly

$$[\texttt{propext},\ \texttt{Classical.choice},\ \texttt{Quot.sound}].$$

The declarations are also registered in `oracle_check.lean`. The focused audit is independently runnable as

```
lake env lean papers/shift-uniform-combes-thomas/ShiftUniformOracle.lean
```

The development contains no `sorry` in the two resolvent modules or the three new root modules.

External author-corpus provenance note. As of August 1, 2026, thirteen corrective or supersession forms concerning other manuscripts had been sent and remained **ENVIADO/PENDIENTE**; none should be resent or described as published while the public records lag. The scope split between 2607.0035 and 2607.0039 remains controlling, and 2607.0089 remains **REVIEW-PENDING**. The complete record-level matrix, promotion rule, and checks are moved to `papers/shift-uniform-combes-thomas/PUBLICATION-PROVENANCE.md`. They are administrative provenance, not hypotheses, evidence, or citations for any mathematical claim in this paper.

11 Conclusion

The useful observation is exact and small: scalar identity shifts disappear from exponential-conjugation defects. Its consequence is structurally larger: a single fixed Combes–Thomas budget controls an entire positive resolvent family with the integrable amplitude $(c+s)^{-1}$. That family is now a machine-checked theorem on the physical gauge-cochain types that the covariance-root interface actually consumes.

The paper-level Stieltjes integral then produces exponential localization of $K^{-1/2}$ with amplitude $2/\sqrt{c}$. This is not advertised as a new result in matrix analysis. Version 0.4 does close one previously open bridge: Lean now constructs the positive real continuous-linear root and derives its exact square, positivity, symmetry, and the corresponding algebraic certificate fields. What remains is narrower and genuinely analytic: identify that root with the Stieltjes integral and transfer the integrated resolvent estimate to the single-bond root-kernel predicate. Until that theorem is formalized, the localized physical certificate remains conditional; the fixed-volume Poincaré wall is unchanged.

References

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